Homework (part 2)

FYSB21

Author
Fredrik Bergelv
fredrik.bergelv@live.se

Lund University, Department of Physics



LUND UNIVERSITY

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Solution Exercise 1

i) Providing the general solution linearly of the two eigenmodes

From the question it was given that

$$m_1\ddot{\phi}_1 = -\frac{m_1 g}{L}\phi_1 + k(\phi_2 - \phi_1) \tag{1}$$
$$m_2\ddot{\phi}_2 = -\frac{m_2 g}{L}\phi_2 - k(\phi_2 - \phi_1). \tag{2}$$

From Equation 1 and Equation 2 one can see that the system is at rest if the left hand side is equal to zero. This gives that $\phi_{1,0} = 0$ and $\phi_{2,0} = 0$. This becomes clearer when the equation is simplified as in Equation 3:

$$\begin{cases} m_1\ddot{\phi}_1 &= -\frac{m_1 g}{L}\phi_1 + k\phi_2 - k\phi_1 \\ m_2\ddot{\phi}_2 &= -\frac{m_2 g}{L}\phi_2 - k\phi_2 + k\phi_1 \end{cases} \implies \begin{cases} m_1\ddot{\phi}_1 &= \left(-k - \frac{m_1 g}{L}\right)\phi_1 + k\phi_2 \\ m_2\ddot{\phi}_2 &= \left(-k - \frac{m_2 g}{L}\right)\phi_2 + k\phi_1 \end{cases} \tag{3}$$

Equation 3 can be rewritten on matrix form:

$$\begin{bmatrix} m_1\ddot{\phi}_1 \\ m_2\ddot{\phi}_2 \end{bmatrix} = \begin{bmatrix} -\left(k + \frac{m_1 g}{L}\right) & k \\ k & -\left(k + \frac{m_2 g}{L}\right) \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix} \tag{4}$$

We provide the ansatz:

$$\begin{bmatrix} \phi_1(t) \\ \phi_2(t) \end{bmatrix} = \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} e^{-i\omega t} \tag{5}$$

This ansatz with Equation 4 gives the following:

$$\begin{bmatrix} m_1 C_1 \\ m_2 C_2 \end{bmatrix} (-\omega^2) e^{-i\omega t} = \begin{bmatrix} -\left(k + \frac{m_1 g}{L}\right) & k \\ k & -\left(k + \frac{m_2 g}{L}\right) \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} e^{-i\omega t}$$
$$\implies \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} (-\omega^2) = \begin{bmatrix} -\left(k + \frac{m_1 g}{L}\right) & k \\ k & -\left(k + \frac{m_2 g}{L}\right) \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix}$$
$$\implies \begin{bmatrix} -m_1\omega^2 & 0 \\ 0 & -m_2\omega^2 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} -\left(k + \frac{m_1 g}{L}\right) & k \\ k & -\left(k + \frac{m_2 g}{L}\right) \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix}$$
$$\implies 0 = \begin{bmatrix} k + \frac{m_1 g}{L} - m_1\omega^2 & -k \\ -k & k + \frac{m_2 g}{L} - m_2\omega^2 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} \tag{6}$$

This is an eigenvalue problem where one can calculate the determinant of Equation 6 to find the oscillation frequencies. We assume $C_1 \neq 0$ and $C_2 \neq 0$.

$$0 = \begin{vmatrix} k + \frac{m_1 g}{L} - m_1\omega^2 & -k \\ -k & k + \frac{m_2 g}{L} - m_2\omega^2 \end{vmatrix} = \left(k + \frac{m_1 g}{L} - m_1\omega^2\right)\left(k + \frac{m_2 g}{L} - m_2\omega^2\right) - k^2$$

$$= k^2 + \frac{k m_2 g}{L} - k m_2\omega^2 + \frac{m_1 g}{L}k + \frac{m_1 m_2 g^2}{L^2} - \frac{m_1 m_2 g}{L}\omega^2 - k m_1\omega^2 - \frac{m_1 m_2 g}{L}\omega^2 + m_1 m_2\omega^4 - k^2$$
$$= m_1 m_2\omega^4 + \left(-k m_1 - k m_2 - \frac{2 m_1 m_2 g}{L} -\right)\omega^2 + \frac{k m_2 g}{L} + \frac{k m_1 g}{L} + \frac{m_1 m_2 g^2}{L^2}$$

$$= m_1 m_2 \left(\frac{g}{L} - \omega^2\right)\left(\frac{g}{L} + \frac{k}{m_1} + \frac{k}{m_2} - \omega^2\right) \tag{7}$$

We solve the polynomial in Equation 7, which gave two solutions $\omega^2_{(i)}$ and $\omega^2_{(ii)}$ as follows:

$$\omega^2_{(i)} = \frac{g}{L} \tag{8}$$

$$\omega^2_{(ii)} = \frac{g}{L} + \frac{k}{m_1} + \frac{k}{m_2} \tag{9}$$

Next step is to calculate the constants C_1 and C_2. This can be done using Equation 6 for $\omega^2_{(i)}$ and $\omega^2_{(ii)}$. $A_{(i)}$ and $A_{(ii)}$ both denote complex constants.

$$\omega^2 = \omega^2_{(i)} \implies 0 = \begin{bmatrix} k + \frac{m_1 g}{L} - \frac{m_1 g}{L} & -k \\ -k & k + \frac{m_2 g}{L} - \frac{m_2 g}{L} \end{bmatrix} \begin{bmatrix} C_1^{(i)} \\ C_2^{(i)} \end{bmatrix} \tag{10}$$

$$\implies 0 = \begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \begin{bmatrix} C_1^{(i)} \\ C_2^{(i)} \end{bmatrix} \implies \begin{bmatrix} C_1^{(i)} \\ C_2^{(i)} \end{bmatrix} = A_{(i)} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\omega^2 = \omega^2_{(ii)} \implies 0 = \begin{bmatrix} k + \frac{m_1 g}{L} - m_1\left(\frac{g}{L} + \frac{k}{m_1} + \frac{k}{m_2}\right) & -k \\ -k & k + \frac{m_2 g}{L} - m_2\left(\frac{g}{L} + \frac{k}{m_1} + \frac{k}{m_2}\right) \end{bmatrix} \begin{bmatrix} C_1^{(ii)} \\ C_2^{(ii)} \end{bmatrix} \tag{11}$$

$$\implies 0 = \begin{bmatrix} -\frac{k m_1}{m_2} & -k \\ -k & -\frac{k m_2}{m_1} \end{bmatrix} \begin{bmatrix} C_1^{(ii)} \\ C_2^{(ii)} \end{bmatrix} \tag{12}$$

$$\implies 0 = \begin{bmatrix} \frac{m_1}{m_2} & 1 \\ 1 & \frac{m_2}{m_1} \end{bmatrix} \begin{bmatrix} C_1^{(ii)} \\ C_2^{(ii)} \end{bmatrix} \implies \begin{bmatrix} C_1^{(ii)} \\ C_2^{(ii)} \end{bmatrix} = A_{(ii)} \begin{bmatrix} m_2 \\ -m_1 \end{bmatrix} \tag{13}$$

When taking into account in both roots of $\omega_{(i)}$ and $\omega_{(ii)}$ we obtain the solutions:

$$\begin{bmatrix} \phi_1^{(i)}(t) \\ \phi_2^{(i)}(t) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \left(A^-_{(i)} e^{-i\omega_{(i)} t} + A^+_{(i)} e^{i\omega_{(i)} t}\right) \tag{14}$$

$$\begin{bmatrix} \phi_1^{(ii)}(t) \\ \phi_2^{(ii)}(t) \end{bmatrix} = \begin{bmatrix} m_2 \\ -m_1 \end{bmatrix} \left(A^-_{(ii)} e^{-i\omega_{(ii)} t} + A^+_{(ii)} e^{i\omega_{(ii)} t}\right) \tag{15}$$

Since we are only interested in the real solutions, since we have a real problem, $A^+_{(i)} = A^{-*}_{(i)}$ and $A^+_{(ii)} = A^{-*}_{(ii)}$. We can introduce the constants B and φ to obtain real constants. We set $A^-_{(i)} = \frac{B_{(i)} e^{i\omega\varphi_{(i)}}}{2}$ and $A^-_{(ii)} = \frac{B_{(ii)} e^{i\omega\varphi_{(ii)}}}{2}$. This gives that we can rewrite the expression above as:

$$\begin{bmatrix} \phi_1(t) \\ \phi_2(t) \end{bmatrix} = \begin{bmatrix} \phi_1^{(i)}(t) \\ \phi_2^{(i)}(t) \end{bmatrix} + \begin{bmatrix} \phi_1^{(ii)}(t) \\ \phi_2^{(ii)}(t) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} B_{(i)} \cos(\omega_{(i)} t - \varphi_{(i)}) + \begin{bmatrix} m_2 \\ -m_1 \end{bmatrix} B_{(ii)} \cos(\omega_{(ii)} t - \varphi_{(ii)}) \tag{16}$$

Answer: $\begin{bmatrix} \phi_1(t) \\ \phi_2(t) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} B_{(i)} \cos(\omega_{(i)} t - \varphi_{(i)}) + \begin{bmatrix} m_2 \\ -m_1 \end{bmatrix} B_{(ii)} \cos(\omega_{(ii)} t - \varphi_{(ii)})$
where $B_{(i),(ii)}$ and $\varphi_{(i),(ii)}$ are free parameters but $\omega_{(i)} = \sqrt{\frac{g}{l}}$ and $\omega_{(ii)} = \sqrt{\frac{g}{l} + \frac{k}{m_1} + \frac{k}{m_2}}$.

ii) Plotting the result

We can plot the angles ϕ_1 and ϕ_2 depending on t. We know that $m_1 = m_2 = 1\,\text{kg}$, $L = 1\,\text{m}$, $k = 1\,\text{N}\,\text{m}^{-1}$ and $\phi_1(0) = 0.1\,\text{rad}$, $\dot{\phi}_1(0) = 0\,\text{rad}\,\text{s}^{-1}$, $\phi_2(0) = 0\,\text{rad}$, $\dot{\phi}_2(0) = 0\,\text{rad}\,\text{s}^{-1}$. Using this we can calculate $B_{(i)}$, $B_{(ii)}$, $\varphi_{(i)}$ and $\varphi_{(ii)}$. Since the angular velocity is equal to zero in $t = 0$ we know $\varphi_{(i)} = 0$ and $\varphi_{(ii)} = 0$. To find $B_{(i)}$ and $B_{(ii)}$ we use use Equation 16 as follows:

$$
\begin{cases}
\begin{bmatrix} 0.1 \\ 0 \end{bmatrix} &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} B_{(i)} \cos(\omega_{(i)} 0) + \begin{bmatrix} m_2 \\ -m_1 \end{bmatrix} B_{(ii)} \cos(\omega_{(ii)} 0) \\
\begin{bmatrix} 0 \\ 0 \end{bmatrix} &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \omega_{(i)} B_{(i)} \sin(\omega_{(i)} 0) + \begin{bmatrix} m_2 \\ -m_1 \end{bmatrix} - \omega_{(ii)} B_{(ii)} \sin(\omega_{(ii)} 0)
\end{cases}
$$

$$
\begin{cases}
\begin{bmatrix} 0.1 \\ 0 \end{bmatrix} &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} B_{(i)} + \begin{bmatrix} m_2 \\ -m_1 \end{bmatrix} B_{(ii)} \\
\begin{bmatrix} 0 \\ 0 \end{bmatrix} &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} 0 + \begin{bmatrix} m_2 \\ -m_1 \end{bmatrix} 0
\end{cases}
\implies
\begin{cases}
\begin{bmatrix} 0.1 \\ 0 \end{bmatrix} &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} B_{(i)} + \begin{bmatrix} m_2 \\ -m_1 \end{bmatrix} B_{(ii)} \\
0 &= 0
\end{cases}
$$

$$
\implies \begin{bmatrix} 0.1 \\ 0 \end{bmatrix} = \begin{bmatrix} B_{(i)} + B_{(ii)} \\ B_{(i)} - B_{(ii)} \end{bmatrix} \implies \begin{cases} B_{(i)} &= 0.05 \\ B_{(i)} &= 0.05 \end{cases}
$$

Since we know that $\omega_{(i)} = \sqrt{\frac{g}{l}} = \sqrt{g}$ and $\omega_{(ii)} = \sqrt{\frac{g}{l} + \frac{k}{m_1} + \frac{k}{m_2}} = \sqrt{g+2}$ we can plot the functions $\phi_1(t)$ and $\phi_2(t)$. The plot of Equation 17 and Equation 18 can be observed in Figure 1 and Figure 2.

$$
\phi_1(t) = 0.05 \cos\left(\sqrt{g} \cdot t\right) + \cos\left(\sqrt{g+2} \cdot t\right) \tag{17}
$$

$$
\phi_2(t) = 0.05 \cos\left(\sqrt{g} \cdot t\right) - \cos\left(\sqrt{g+2} \cdot t\right) \tag{18}
$$

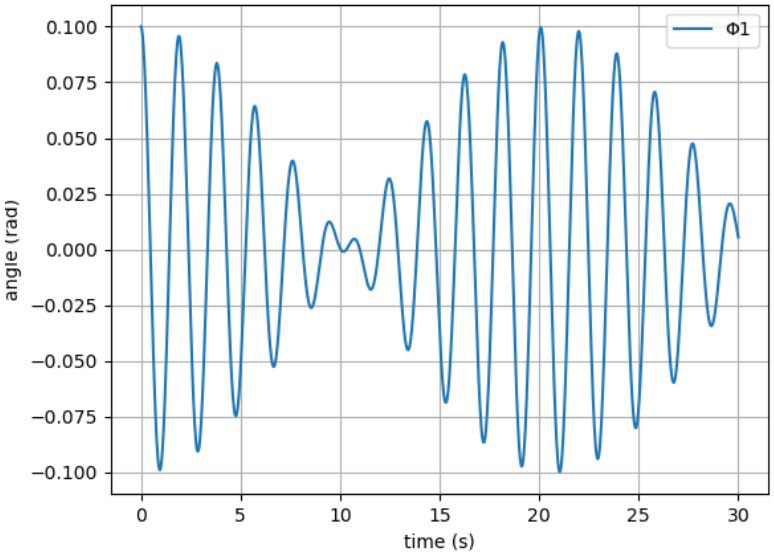


Figure 1: The position of ϕ_1 depending on t.

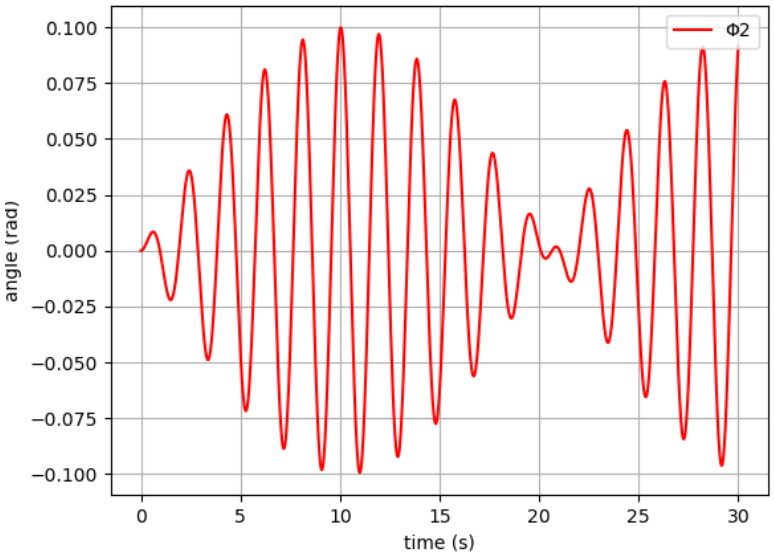


Figure 2: The position of ϕ_2 depending on t.

Beating is the interference of waves of different frequencies. The resulting interference pattern produces a periodic variation known as beats. This can be calculated by adding the two different functions, in this scenario that is ϕ_1 and ϕ_2. The frequency of the beat is the difference in frequencies of the two functions, resulting in a *perceived* change in volume with that frequency. From the calculations below and Figure 3 one can tell that we have a beating of 0.5 Hz in this case.

$$\begin{aligned}\phi_1(t) + \phi_2(t) &= 0.05\cos\left(\sqrt{g}t\right) + \cos\left(\sqrt{g+2k}t\right) + 0.05\cos\left(\sqrt{g}t\right) - \cos\left(\sqrt{g+2k}t\right) \\ &= 0.1\cos\left(\sqrt{g}t\right) \implies f_{beat} = \frac{\sqrt{g}}{2\pi} \approx 0.5\,\text{Hz}\end{aligned}$$

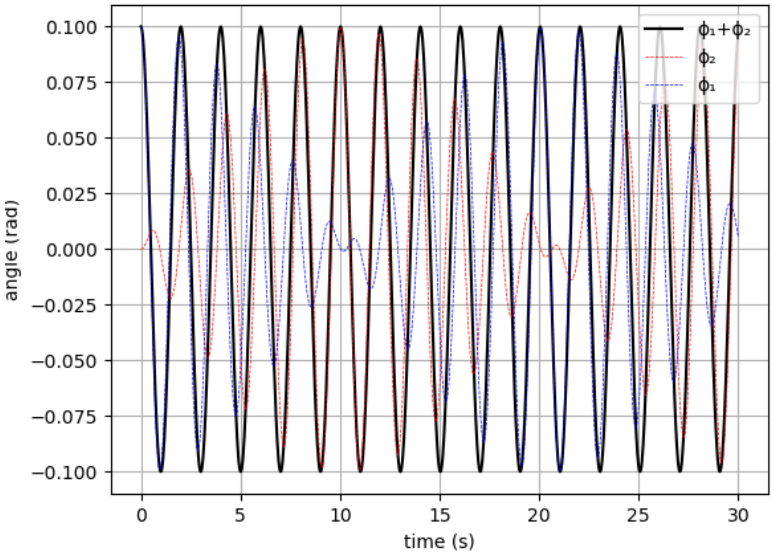


Figure 3: Plotting of the interference $\phi_1 + \phi_2$ showing the beating with its frequency.

Solution Exercise 2

How carbon dioxide and methane leads to global warming

Molecules are a classic example of systems with couple oscillations. Thus, the frequency of the coupled oscillations can be calculated like in Exercise 1. If one calculates these frequencies and evaluate the ones which are non zero, one obtains the molecules different vibrational frequencies. A molecule with a vibrational frequency may result in that molecule being a oscillating electric dipole. Gases as oxygen and nitrogen do not have any dipole moment since they are composed of one type of atom, but gasses such as carbon dioxide and methane do have dipole moment. Electric dipoles that oscillates exhibit a pronounced interaction with electromagnetic waves of the same frequency, which result in absorption of radiation. Thus, a vibrating molecule of a specific frequency will absorb radiation of that frequency. These frequencies will be different for different molecules. The vibrational frequency can be calculated to a vibrational wavelength, this can be viewed in Figure 4.

The earth is a system witch receives energy from incoming light from the sun and absorbs some of that light. This results in the earth obtaining a temperature witch in turn leads to radiation. The earth is a black-body of about 300 K, which means that it will spit out electromagnetic radiation with a frequency of 18 THz [1]. This frequency can be calculated to a wavelength of 10 µm, which can be seen in Figure 5. When comparing this with one of the vibrational wavelength's of the carbon dioxide atoms we can see that the carbon carbon dioxide has a vibrational wavelength around 14 µm (Figure 4). The same can be observed for the methane molecule with a vibrational wavelength around 8 µm (Figure 4). If one observes Wien's displacement law one can see that the outgoing radiation of the earth will be partially blocked by both carbon dioxide and methane. This can be viewed for carbon dioxide in Figure 5. But if one looks at the incoming radiation from the sun, with a surface temperature of 5700 K one can see that this is not blocked by carbon dioxide nor methane [2]. It is therefor clear that an increase in carbon dioxide and methane will not block incoming radiation from the sun, but it will block outgoing radiation from the earth. This increase will thus lead to a shift in the energy balance of the earth, meaning more energy would stay and less energy would leave resulting in a higher global temperature.

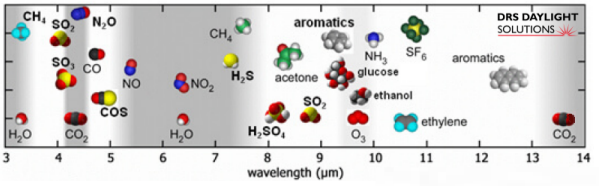


Figure 4: Molecules and some of their vibrational wavelength's [2].

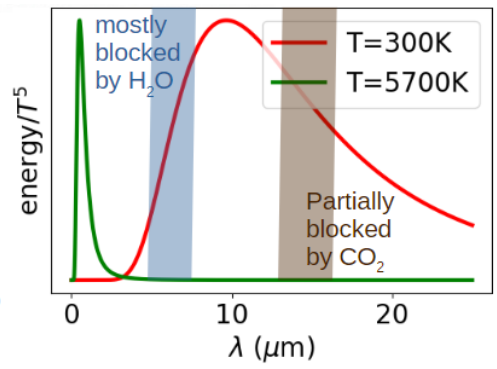


Figure 5: Earths absorption and radiation with the wavelengths of carbon dioxide and water [2].

Solution Exercise 3

i) Determining the spatial profile at t=0

In equation

$$h(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} dk \Re\left\{a(k)e^{i(kx-\omega(k)t)}\right\}, \tag{19}$$

We know that $k_0 = \frac{2\pi}{5\,\mathrm{m}}, \alpha = \frac{k_0}{5}$, and $C = 0.1\,\mathrm{m}$. it is also known that $a(k) = 2\pi h_0\delta(k) + \frac{2\alpha C}{(k-k_0)^2+\alpha^2}$. This gives the equation:

$$h(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} dk \Re\left\{\left(2\pi h_0\delta(k) + \frac{2\alpha C}{(k-k_0)^2+\alpha^2}\right)e^{i(kx-\omega(k)t)}\right\} \tag{20}$$

Since we that $t = 0$ know Equation 20 can be rewritten as:

$$\begin{aligned} h(x,0) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} dk \Re\left\{\left(2\pi h_0\delta(k) + \frac{2\alpha C}{(k-k_0)^2+\alpha^2}\right)e^{ikx}\right\} \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} dk \Re\left\{2\pi h_0\delta(k)e^{ikx} + \frac{2\alpha C e^{ikx}}{(k-k_0)^2+\alpha^2}\right\} \\ &= \frac{1}{2\pi}\Re\left\{\int_{-\infty}^{\infty} dk\left[2\pi h_0\delta(k)e^{ikx} + \frac{2\alpha C e^{ikx}}{(k-k_0)^2+\alpha^2}\right]\right\} \\ &= \frac{1}{2\pi}\Re\left\{\int_{-\infty}^{\infty} dk 2\pi h_0\delta(k)e^{ikx} + \int_{-\infty}^{\infty} dk\frac{2\alpha C e^{ikx}}{(k-k_0)^2+\alpha^2}\right\} \end{aligned} \tag{21}$$

We have now obtained two integrals where the one on the left can be solved. The one on the right we set $u = k - k_0$, which implies that $k = u + k_0$ and $k_0 = k - u$, which makes it solvable.

$$\begin{aligned} &= \frac{1}{2\pi}\Re\left\{\int_{-\infty}^{\infty} dk 2\pi h_0\delta(k)e^{ikx} + \int_{-\infty}^{\infty} du\frac{2\alpha C e^{i(u+k_0)x}}{(u^2+\alpha^2)}\right\} \\ &= \frac{1}{2\pi}\Re\left\{\int_{-\infty}^{\infty} dk 2\pi h_0\delta(k)e^{ikx} + e^{ik_0x}\int_{-\infty}^{\infty} du\frac{2\alpha C e^{iux}}{(u^2+\alpha^2)}\right\} \end{aligned}$$

We have now obtained to integrals which are solvable.

$$\begin{aligned} &= \frac{1}{2\pi}\Re\left\{2\pi h_0 e^0 + e^{ik_0x}C2\pi e^{-\alpha|x|}\right\} \\ &= h_0 + C\cos(k_0x)\,e^{-\alpha|x|} \end{aligned}$$

If we plug in our values for $k_0,\ \alpha,\ C$ we obtain the wave function (in meters) to be:

$$h(x,0) = h_0 + 0.1\cos\left(\frac{2\pi}{5}x\right)e^{-\frac{2\pi}{25}|x|} \tag{22}$$

One can tell from Equation 22 that the largest value of $h(x, t = 0)$ is near $x = 0$, from where it gets smaller exponentially in both directions periodically. To get a better feeling for the range we can plot this. From Figure 6 one can say that substantial values a roughly $x \in [-15, 15]$. But if we only evaluate point which are half the peak we obtain values $x \in [-0.74, 0.74]$.

If we want to identify $\tilde{h}(k, t = 0)$ we can see from Equation 21 that it is written like the inverse Fourier transform. One can therefor conclude that $a(k) = \tilde{h}(k, t = 0)$, which gives the equation:

$$\tilde{h}(k, 0) = 2\pi h_0 \delta(k) + \frac{2\alpha C}{(k - k_0)^2 + \alpha^2} \tag{23}$$

From Equation 23 we can tell that we will have a large, but very narrow, peak at $k = 0$. We can also see that we will have a maximum around $k = k_0$. To see the range we can plot this function. From Figure 7 one can say that substantial values a roughly $k \in \{0, [0.5, 2]\}$. But if we only look at values which are half the peak we obtain that $k \in \{0, [1, 1.5]\}$.

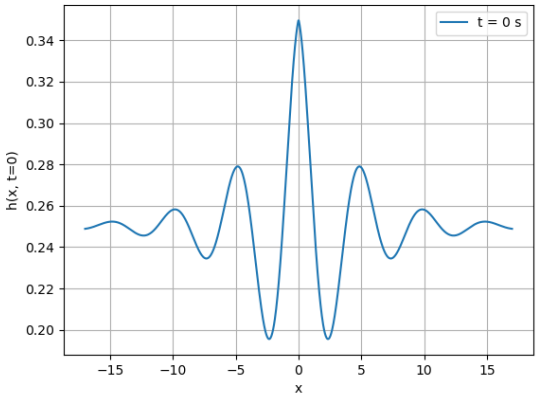


Figure 6: The calculated function $h(x, t = 0)$, where $h_0 = 0.25\,\text{m}$.

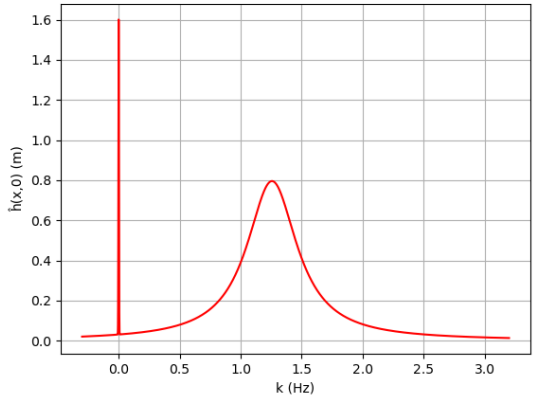


Figure 7: The identified function $\tilde{h}(k, t = 0)$, where $h_0 = 0.25\,\text{m}$.

Answer: The spatial function at time zero is $h(x, 0) = h_0 + 0.1 \cos\left(\frac{2\pi}{5} x\right) e^{-\frac{2\pi}{25}|x|}$. Where substantial values are roughly between $x \in [-15, 15]$. If we viewing $\tilde{h}(x, 0)$ we can tell that substantial values are roughly $k \in \{0, [0.5, 2]\}$. But if we only look at values which are half the peak we obtain that $x \in [-0.74, 0.74]$ and $k \in \{0, [1, 1.5]\}$.

ii) Determining the evolution in time for a shallow water wave

We can calculate the evolution in time, if we approximate $k \approx |k|$, as follows:

$$
\begin{aligned}
h(x,t) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} dk \Re\left\{\left(2\pi h_0\delta(k) + \frac{2\alpha C}{(k-k_0)^2+\alpha^2}\right)e^{i(kx-\omega(k)t)}\right\} = ... \\
&= \frac{1}{2\pi}\Re\left\{\int_{-\infty}^{\infty} dk 2\pi h_0\delta(k)e^{ikx-i\sqrt{gh_0}kt} + e^{ik_0x}\int_{-\infty}^{\infty} du \frac{2\alpha C e^{iux}e^{-i\sqrt{gh_0}(u+k_0)t}}{(u^2+\alpha^2)}\right\} \\
&= \frac{1}{2\pi}\Re\left\{\int_{-\infty}^{\infty} dk 2\pi h_0\delta(k)e^{ikx-i\sqrt{gh_0}kt} + e^{k_0\left(ix-i\sqrt{gh_0}t\right)}\int_{-\infty}^{\infty} du \frac{2\alpha C e^{u\left(ix-i\sqrt{gh_0}t\right)}}{(u^2+\alpha^2)}\right\} \\
&= \frac{1}{2\pi}\Re\left\{2\pi h_0 + e^{ik_0\left(x-\sqrt{gh_0}t\right)}\int_{-\infty}^{\infty} du \frac{2\alpha C e^{ui\left(x-\sqrt{gh_0}t\right)}}{(u^2+\alpha^2)}\right\} \\
&= h_0 + \frac{1}{2\pi}\Re\left\{e^{k_0i\left(x-\sqrt{gh_0}t\right)}\int_{-\infty}^{\infty} du \frac{2\alpha C e^{iu\left(x-\sqrt{gh_0}t\right)}}{(u^2+\alpha^2)}\right\} \\
&= h_0 + \frac{1}{2\pi}\Re\left\{e^{k_0i\left(x-\sqrt{gh_0}t\right)}2\pi\alpha C e^{-\alpha\left|x-\sqrt{gh_0}t\right|}\right\} \\
&= h_0 + C\cos\left(k_0(x-\sqrt{gh_0}t)\right)e^{-\alpha\left|x-\sqrt{gh_0}t\right|} \quad \text{We plug in values for } C, \alpha, k_0 \text{ and } h_0 \\
&= 0.25 + 0.1\cos\left(\frac{2\pi}{5}(x-\sqrt{0.25g}t)\right)e^{-\frac{2\pi}{25}\left|x-\sqrt{0.25g}t\right|}
\end{aligned}
$$

We can calculate the group velocity for $\omega_{disp} = \sqrt{gh_0}|k| \approx \sqrt{gh_0}k$ as follows:

$$v_g = \frac{\mathrm{d}\omega_{disp}(k)}{\mathrm{d}k} = \sqrt{gh_0} \tag{24}$$

If we plug in $h_0 = 0.25\,\mathrm{m}$ and $g = 9.82\,\mathrm{m\,s^{-2}}$ into Equation 24 we obtain that the group velocity $v_g \approx 1.6\,\mathrm{m\,s^{-1}}$.

The phase can be calculated as follows:

$$v_p = \frac{\omega_{disp}(k)}{k} = \frac{\sqrt{gh_0}|k|}{k} \approx \sqrt{gh_0} \tag{25}$$

As before we plug in h_0 and g into Equation 25 to obtain that the phase velocity $v_p \approx 1.6\,\mathrm{m\,s^{-1}}$. One can tell from these calculations that since the velocities are identical, we should obtain a wave packet which does not change shape over time.

Answer: The function in meters can be described as $h(x,t) = 0.25 + 0.1\cos\left(\frac{2\pi}{5}(x-\sqrt{0.25g}t)\right)e^{-\frac{2\pi}{25}\left|x-\sqrt{0.25g}t\right|}$. This corresponds with our obtained $h(x, t=0)$. For this function, the group and phase velocity is $v_g \approx 1.6\,\mathrm{m\,s^{-1}}$ and $v_p \approx 1.6\,\mathrm{m\,s^{-1}}$.

iii) Solving the general formula for h(x,t) numerically in shallow water

When solving this integral numerically, Figure 8 was obtained. When calculating v_g and v_p we obtained that these where equal. This results in the the wave packet not changing form when moving, which almost seems to be the case in Figure 8. We can see that the peaks get a little lower when t increases.

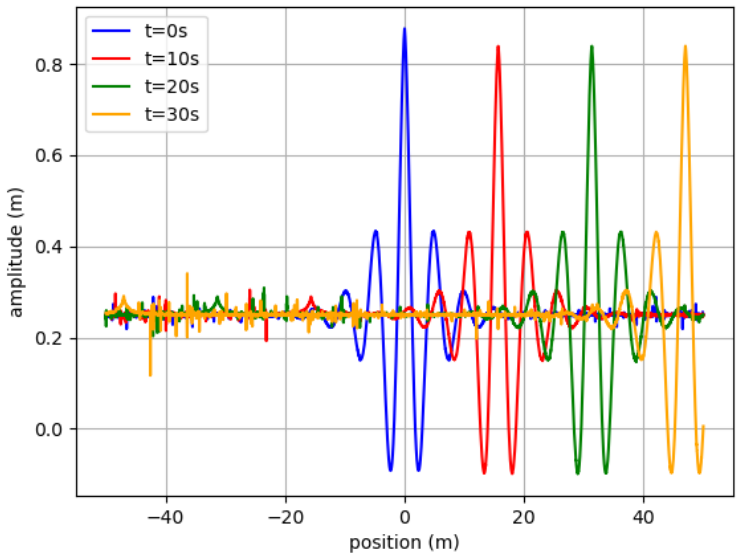


Figure 8: The function $h(x,t)$ solved numerically for different times and $h_0 = 0.25\,\mathrm{m}$.

When viewing Figure 9 we can tell that our approximation actually changes the plot slightly. One can see from Figure 8 that our wave function does change form depending on t, but this does not happen with the approximations as seen in Figure 9. This deviation occurs because we assumed $|k| = k$, but if we instead would calculate the group velocity v_g for $v_g = \omega_{disp} = \frac{\mathrm{d}\sqrt{gh_0}|k|}{\mathrm{d}k}$ we would get a group velocity depending on k. This is exactly what happens when we plot without the approximation in Figure 8.

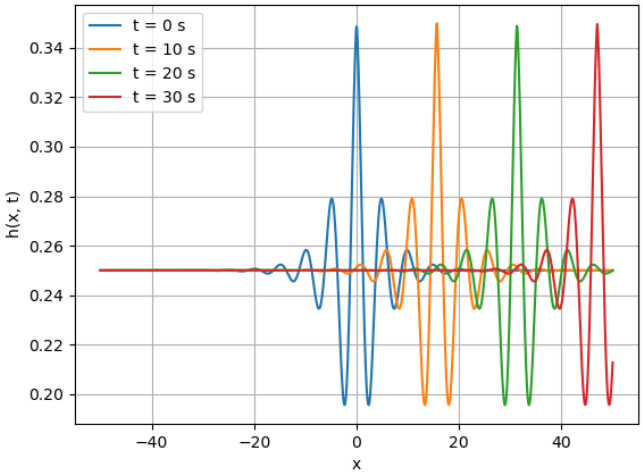


Figure 9: A plot of the function $h(x,t)$ where and $h_0 = 0.25\,\mathrm{m}$. Here we plot with the approximation that $|k| = k$.

iv) Determining the evolution in time for a deep water wave

When we have deep water the dispersion relation is $\omega_{disp} = \sqrt{g|k|}$. Thus, the group velocity can be calculated as:

$$v_g = \frac{\mathrm{d}\omega_{disp}(k)}{\mathrm{d}k} = \frac{\mathrm{d}\sqrt{g|k|}}{\mathrm{d}k} = \frac{1}{2}\frac{g|k|}{\sqrt{g|k|}k} = \frac{\sqrt{g}k}{2|k|^{\frac{3}{2}}} \tag{26}$$

Using Equation 26 and k_0 we obtain that $v_{g,0} \approx 1.4\,\mathrm{m\,s^{-1}}$.

The phase velocity can be calculated as:

$$v_p = \frac{\omega_{disp}(k)}{k} = \frac{\sqrt{g|k|}}{k} \tag{27}$$

Using Equation 27 and k_0 we obtain that $v_{p,0} \approx 2.8\,\mathrm{m\,s^{-1}}$. Since Equation 24 and Equation 25 are not the same, we would obtain a wave packet which changes shape over time.

Answer: The group and phase velocity changes depending on k which can bee seen in Equation 26 and Equation 27. The velocities at $k = k_0$ was calculated to $v_{g,0} \approx 1.4\,\mathrm{m\,s^{-1}}$ and $v_{p,0} \approx 2.8\,\mathrm{m\,s^{-1}}$.

v) Solving the general formula for h(x,t) numerically in deep water

When changing the depth h_0 Figure 10 was obtained.

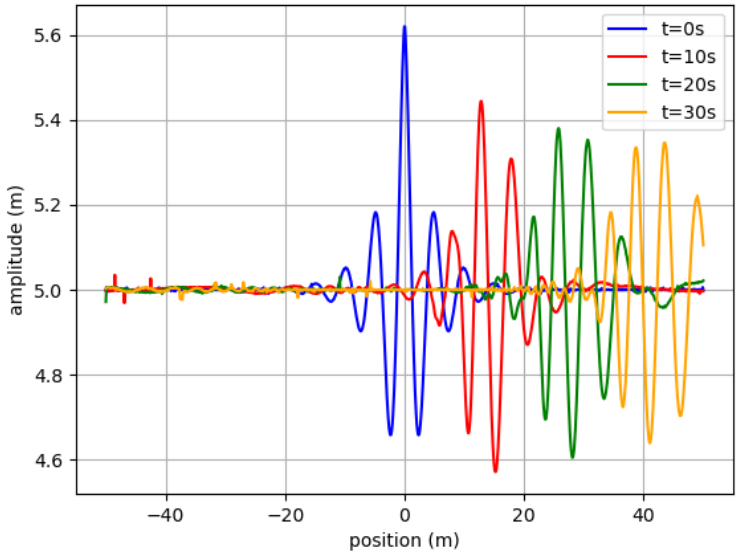


Figure 10: The function $h(x,t)$ solved numerically for different times and $h_0 = 5\,\mathrm{m}$.

References

[1] Andreas Wacker. Compendium mathematical methods for vibrations, waves, and diffusion. *Lund University, Mathematical Physics*, page 58, 2023.

[2] Andreas Wacker. Coupled oscillations. `https://canvas.education.lu.se/courses/24791/files/4329471?module_item_id=1030361`, *Lund University, Science Faculty, Mathematical Physics,* slide 32, 2023.